

Representation of Expressions

1 Expressions

- operators and operands of expressions
- expression trees are not binary

2 Evaluation

- numeric and symbolic evaluation
- binary expression trees for numerical evaluation

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Introduction to Symbolic Computation
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Operators and Operands of Expressions

A polynomial p with m terms, n variables, coefficients in K :

$$p = \sum_{i=1}^m c_j \prod_{j=1}^n x_j^{d_{i,j}} \in K[x_1, x_2, \dots, x_n].$$

Three operators:

- 1 $\sum_{i=1}^m$ represents addition, for i from 1 to m .
- 2 $\prod_{j=1}^n$ represents multiplication, for j from 1 to n .
- 3 The exponentiation is in the raised superscripts, in $x_j^{d_{i,j}}$, $d_{i,j}$ is the power of x_j in the i -th monomial.

Exponentiation is binary, addition and multiplication have operand *lists*.

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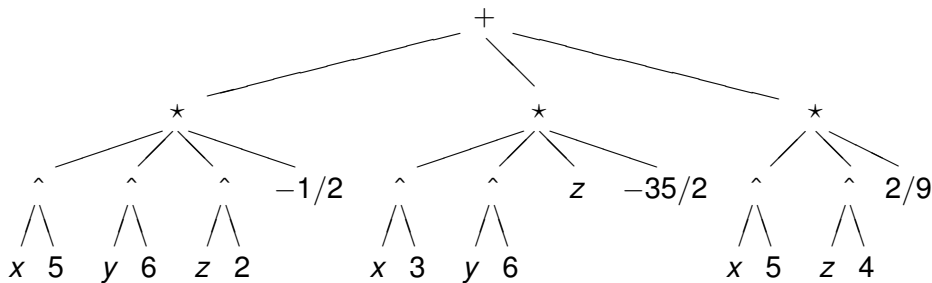
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Expression Trees are not Binary

$$p = -\frac{1}{2}x^5y^6z^2 - \frac{35}{2}x^3y^6z + \frac{2}{9}x^5z^4$$



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numeric and symbolic evaluation

We distinguish between numeric and symbolic evaluation.

- Numerical evaluation:

- ▶ The variables in an expression are replaced by numbers.
- ▶ The result is a number.

- Symbolic evaluation:

- ▶ The variables in an expression are replaced by symbols.
- ▶ The result is an expression.

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binary expression trees for numerical evaluation

The numerical arithmetical operations are binary.

Therefore, for numerical evaluation, the expression trees are binary.

For example, $-\frac{1}{2}x^5y^6z^2$ has as expression tree:

